

Action Items and Lessons from ChE Retreat

1. Inculcating student focused approach
 - a. During advising, ask students to share their CVs.
 - i. **Policy modification:** Requirement for students to share their CV
 - b. Encourage UG students to volunteer in research during summer. Lower down expectations for outcomes. Have them attend group meetings.
 - i. **Policy Discussion:** 4 UG student required to be associated with each lab?
 - c. Determine: Add a CHE 499 lecture on CV writing and/or AIChE student chapter meeting on CV writing/interview skills (with ChE professors)
 - d. **Determine:** Ways to support students via summer professional development program?
 - e. **Determine:** How to have paid internships of students in faculty startups?
 - f. **Promote:** Summer internships programs like REU; Engineering Study Abroad, OPT/CPT, Co-op programs.
 - g. Promoting extracurricular student activities- one-on-one guidance, don't just announce in class- follow-up with individual students (Betul comments)
 - h. Computational assignments with software (Michael comments)
 - i. Introducing undergraduates to research labs (Vivek comments); make initial commitment very flexible, low-level of participation, attend lab meetings, perform simple tasks, etc.
 - j.
2. Undergraduate Program
 - a. **Action:** Discuss within the UG committee
 - i. Create a matrix of courses that students can choose from to emphasize their interest in different skill sets or specializations in ChE ("Lego" course planning)
 - ii. Reduce 3 credit hours from curriculum. Some options
 1. Remove CME 260 or make CHE 230/CME 260 a Selective Option
 2. Make CHEM 342 (PChem1) and CHEM 222 Analytical Chemistry a Selective Option
 3. Remove ECE 210
 - iii. Add one credit to 205 and include Aspen (this will help students wanting concentrations).
 - iv. Add "Industrial Processes in Chemical Engineering"
 - v. Add "Statistical Methods" course

- vi. Add “Data Science” courses
 - vii. Add topics for “Introduction to Chemical Engineering and Sustainability”
 - 1. Include AI and Python
 - 2. Ethics
- 3. Recruitment
 - a. Create a list of recruitment events and share with faculty and keep track of faculty that participate in the events to
- 4. Graduate Program
 - a. **Action:** Addressing the sanction by August
 - b. **Policy:** Streamlining Prelim (within first 5 semesters) and PhD procedure
 - i. **Policy:** Screening methodology to move PhD candidates to MS.
 - c. **Action:** PI/Student agreement finalization
 - d. **Policy:** Streamlining process for conflict resolution between PI and student.
 - e. **Action:** Ensure one seminar per semester by graduate school representative
 - f. **Determine:** Reduction of core courses (or make them selective) and increase electives.
 - g. **Determine:** Strategies to recruit from diverse places
 - h. **Action:** Ensure PhD fellowship application deadlines are met
 - i. **Policy:** Ensure there is AI Statements and citations in thesis
- 5. Other Improvements
 - a. Updating Unit Operations
 - i. Reaction Engineering Module
 - 1. **Action:** Purchase an experimental unit
 - b. **Action:** Categorize student’s AIChE visits into three categories based on their activities with faculty advisor/head oversight on choosing students
 - i. Chem-E-Car (5 students)
 - ii. CUBE (5 students)
 - iii. Research Presentations (5 students)
 - c. Institute “ChE Alumni Association”
 - i. **Action:** Work with COE to establish Alumni Association.
 - d. **Action:** Establish “Industry Engagement Committee”
 - i. Incorporation of industry into coursework
 - ii. Hackathons/Challenges/projects with industrial sponsors
 - iii. Student visits to
 - 1. Abbvie

2. Kraft
 3. Amgen
 4. Other- (Michael discussion)
- e. **Action:** Form a committee to determine faculty awards that we can apply
 - i. Update ChE website with faculty accomplishments/awards (Meghan King)
6. Prepare for ABET
 - a. **Actions**
 - b. Summer 2025:
 - i. Gather information- Box folder
 - ii. Course information- Update Learning outcomes, performance indicators, upload student output, ABET syllabi
 - iii. Faculty course rubric assessments
 - c. Fall 2026
 - i. Gather information for Self-Study Report
 - ii. Perform assessment analyses
 - iii. Faculty information
 - iv. Continue course rubric assessments-faculty
 - v. Continue updating course information-faculty
 - d. Spring 2026
 - i. Gather Department, College, University, non-ChE course information
 - ii. Self-study report deadline ~ June 2026
 - e. Summer 2026
 - i. Modify and update ABET Review presentation system
 - ii. Additional updates to Self-study report
 - f. Fall 2026
 - i. Finalize presentation system
 - ii. Plan for ABET reviewer visit
 - g. ABET review; reviewer and date TBD
7. **Action:** Use of AI
 - a. ChE Department policy; How to address in courses
 - i. Betul comments on AI programs and their effectiveness
 - b. ABET policy?
 - c. Computer Aids in ChE (CACHE)
 - i.
 - d.

